

Austin Jones

ADDRESS: 101 E 5th Ave Apt. D-304, Knoxville, TN – PHONE: 615-962-3732

EMAIL: ajones53.aj@gmail.com – GITHUB: <https://github.com/ajone239>

Objective: Software Engineer with experience in Embedded Software and Cloud Development looking for cool problems to solve.

Skills

Working Knowledge: Xilinx FPGAs, Xilinx MPSoCs, Petalinux, Intel FPGAs, POSIX
Professional Languages: C++, C, C#, Python, Shell, VHDL
Recreational Languages: Rust, Haskell, BF, Groff
Office Skills: Excel, GIMP, Visio, L^AT_EX

Work Experience

CURRENT - JAN 2024	Uster - Knoxville, TN Work to make applications and solutions to facilitate Textile Grading devices Software Engineer: Jan 2024 - Current • Own an application for aggregating and statusing state across large sites. – Own the project from customer spec to delivery of 1.0. – Design a UI/UX to show 500 parameters from 50 machines. – Semi-real time updates from 50 machines with smooth UI. – Develop lab infra to allow for a swarm of virtual devices to exercise the application away from customer site. – Accidentally write a UI framework. * Converts a JSON UI spec into WPF screens. * Allows for easy maintenance of 50 ui screens and hundreds of files. * Consolidates field validation, styling, and data binding per field type.. • Inherit and support utility software for manufacture of products. – Solely inherit a 5 year old application from another team of 5. – Support its use on site for ongoing manufacture of company products. – Assess work needed to go from MVP hybrid use to full use. – Own this work and its pre-release validation. – Over see the validation and deployment of the new procedure. • Provide software support for implementing research algorithms into production software. • Developed algorithms to align data product across instruments. • Organize and run efforts relating to company purpose and employee development. – Spearhead a grass roots program for employee recognition. – Start a group to help employees gain autonomy for projects they want to happen. • Take on a summer intern to get an intern program started. – Plan and manage work of the intern. – Learn how to let people fail to learn. – Organize lunch and learns, knowledge transfer sessions, and general intern things.
-----------------------	--

NOV 2023 - JAN 2022	PolySign - Remote Collaborated with many teams across multiple products and technologies to build out features and products for a Digital Asset Custodian in the quickly evolving Cryptocurrency market. Senior Software Engineer: Jan 2023 - Nov 2023 - Contributed to Escrow feature of Python microservice backend to meet new customer requirements: - Redesigned feature data model and control flow to increase robustness. - Designed a new data flow to allow for more flexibility in the feature. - Implemented an end-to-end POC for an unreleased feature across multiple projects. - Worked on legacy C/Rust codebase to support features in other products. - Performed monthly Dev-on-Call rotations debugging live issues across multiple environments including: dev, testing, pre-prod, and prod. - Planned, hosted, and participated in a company wide Hackathon. - Organized and hosted a weekly engineering reading club. Software Engineer: Jan 2022 - Dec 2022 - Helped stand up a green field project in Rust aimed at performing fast, distributed multiparty compute signing: - Designed fully asynchronous architecture for distributed message signing increasing efficiency and robustness. - Overhauled network stack to use Tokio's Tonic gRPC over VirtioVsock Sockets increasing dev velocity and code cleanliness. - Redesigned CI/CD Pipeline for multi-repo project to minimize redundancy in builds, regression testing, and integration testing. - Integrated with main product as a POC.
AUG 2021 - MAY 2021	Garmin International - Olathe, KS <i>Embedded Software Intern</i> Developed code for microcontroller to control power on logic for an unreleased product. - Utilized MCU to monitor main board status and determine the power state of the board. - Referenced PCB schematics and MCU documentation to integrate software and hardware. - Gained strong familiarity with ARM/STM microcontrollers.
NOV 2021 - JAN 2019	Siemens Molecular Imaging - Knoxville, TN <i>Electrical R&D Intern</i> Conduct projects, both individually and collaboratively, to provide value to the ER&D team: - Presented findings from work to a large technical audience to demonstrate the validity of a new system architecture. - Long term projects: - Utilize a Xilinx MPSoC to test limitations of ARM Core processing vs an FPGA implementation to assess cost reductions. - Used multiple Raspberry Pis with off the shelf networking hardware to conduct data path tests and compare the bandwidth with current custom solutions. - Work with a team of engineers using GitLab to write various driver code for a custom Linux image. - Developed Firmware, Embedded Software, and Application Software for a unit test fixture that is used during production of PET electronics.

Personal Projects

JAN 2025	ABC Notebook An app that is meant to help with a CBT journaling style. Made in React Native with the Expo Router. Built and deployed with EAS.
DEC 2024	My blog A place for me to put posts and articles. I mainly write about personally philosophy. Made with Svelte 5.
MAY 2024	Austin's Arcade A collection of JavaScript games. They are mostly remakes of classic arcade games. They've been an exercise in me working fast and shipping.
DEC 2023	Connect Four Bot A connect four bot written in Rust. It implements basic alpha-beta minimax. At a shallow depth it's formidable - deeper depths it is impossible.
JAN 2021	Light Sign with Web Server Control Controlled an LED Array to display text specified on a web page. Custom web server written in Rust hosted control page and sent data over UART. Arduino received UART data and displayed data on LED Array.

Education

University of Tennessee Knoxville		Anticipated Graduation: Dec 2021
Bachelor of Science in COMPUTER ENGINEERING		GPA: 4.0/4.0
<i>Current</i> AUG 2019	Engineering Professional Practice Office - University of Tennessee <i>Student Ambassador</i> Assist UT engineering students to improve resumes, interview/inter-person skills, and overall professional skill. Work with other ambassadors to facilitate networking vents and career fairs.	
JAN 2021	Light Sign with Web Server Control - Personal Project Controlled an LED Array to display text specified on a web page.	
APR 2019	Password Manager Browser Extension - University of Tennessee <i>Data Structures and Algorithms Final Project</i> Collaborated with a group of peers using Github to create a functional Chrome extension that manages a user's passwords for multiple sites.	
APR 2019	Render Component for Firmware Based Pong - University of Tennessee <i>Digital Systems Design Final Project</i> Reimplemented the retro game Pong on an FPGA using VHDL. Component was implemented to read the game state and render all game objects and scores.	

Scholarships and Honors

MAY 2019	Gonzalez Family Awards for Outstanding Computer Engineering Junior
AUG 2019 - <i>Current</i>	S. T. Harris Scholarship
AUG 2019 - <i>Current</i>	Herschel C. and Louise Runnion Brand Engineering Scholarship